

PREPARED FOR: TREVOR KNIGHT, GMD · M4CLEANTECH LLC INTERNAL FINANCE BRIEF

M4CleanTech Chester

AD/RNG Primary · Tire BOO Secondary · Land Control via City Ground Lease

The detailed financial brief for M4CleanTech's Chester PA deployment — Anaerobic Digestion / Renewable Natural Gas as the bankable, community-positive primary anchor; Klean/KBR TyreNova BOO as a capital-free secondary revenue stream. Ground lease from City/DCIDA provides site control without acquisition capital. Revenue-first structure.

M4CleanTech LLC · Merge 4 Design LLC · M4Di™ · Aubyn Architecture LLC

SECTION 1 — M4CLEANTECH CHESTER REVENUE AT STEADY STATE

What M4CleanTech Earns from Chester — All Streams

Revenue first. M4CleanTech Chester has three distinct revenue streams, each flowing from a different component of the ground lease structure. Together they represent \$4.5M–\$8.2M gross per year at Year 3–5 steady state, plus a one-time developer fee at BOO financial close. All figures ROM ±30–40%.

\$3.5M– \$6.9M AD/RNG Facility — Gross Annual Tipping fees \$1.8M–\$3.2M + D3 RINs \$1.4M–\$3.1M + digestate \$300K–\$600K	\$150K– \$300K Tire BOO Ground Lease Spread Klean/KBR sublease net of nominal City rent · Zero M4CT capital at risk	\$500K– \$900K AD/RNG Developer Fee + M4Di FeasOps PSA developer fee at AD/RNG financial close + M4Di P3 engagement \$22K	\$4.15M– \$8.1M TOTAL GROSS · M4CleanTech Chester Annual recurring (ex developer fee) at Year 3–5 steady state · ROM ±30–40%
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Revenue Stream Detail

REVENUE STREAM	DESCRIPTION	ROM AMOUNT	TIMING	RISK PROFILE
AD/RNG — Tipping Fees	Regional food waste haulers, FOG processors, institutional waste streams. Competitive rate \$60–\$80/ton. Signed tipping contracts are the primary underwriting input for CDFI lender.	\$1.8M–\$3.2M/yr	Year 1 ops	Contract-based · stable

REVENUE STREAM	DESCRIPTION	ROM AMOUNT	TIMING	RISK PROFILE
AD/RNG — D3 RIN Credits	Federal RFS D3 (cellulosic) RIN credits for food-waste-derived RNG. 11.7 D3 RINs per MMBtu RNG at \$2–\$4/gallon equivalent market price. Stress-test at \$1.50 floor.	\$1.4M–\$3.1M/yr	Year 1 ops	Market-priced · D3 floor risk
AD/RNG — Digestate Sales	Nutrient-rich soil amendment from AD process. Regional ag and compost buyer network. Secondary but reliable; not credit-sensitive.	\$300K–\$600K/yr	Year 2+	Offtake contracts
AD/RNG GROSS TOTAL	Before operating expenses. NOI estimate after ops: \$1.2M–\$2.8M/yr at Year 3. DSCR on \$4M–\$8M CDFI debt at 4–6%: 1.8×–3.5×.	\$3.5M–\$6.9M/yr	Year 2–3	Strong DSCR
Tire BOO — Ground Lease Spread	City charges M4CT nominal rent (\$1–\$5K/yr). Klean/KBR pays M4CT market industrial ground rent on BOO parcel. Spread is net revenue. Zero capital from M4CT.	\$150K–\$300K/yr	Year 2 BOO ops	BOO structure — no capital risk
Tire BOO — Retained Equity	5–15% equity stake in Klean/KBR DE BOO facility negotiated at deal formation. Realized at refinancing or facility sale.	5–15% facility value	Exit Year 8–15	Long-term upside
AD/RNG Developer Fee (one-time)	M4CT bills AD/RNG development partner (Bioenergy Devco / RNG Energy Solutions) for site development services under PSA. Recoverable at financial close.	\$480K–\$880K	At AD/RNG fin. close	One-time · covers predevelopment
M4Di FeasOps P3	M4Di Phase 1A predevelopment engagement on the waterfront site. Billed to project at \$22,000. Establishes G0 ROM and kicks off grant applications.	\$22,000	Immediate	M4Di standard fee
TOTAL — M4CleanTech Chester (Annual Recurring)	AD/RNG gross + Tire BOO spread. Developer fee excluded (one-time). M4CT retains 100% of AD/RNG NOI + 100% of BOO spread.	\$3.65M–\$7.2M/yr	Year 2–3+	ROM ±30–40%

ROM ±30–40%. AD/RNG NOI of \$1.2M–\$2.8M/yr net assumes 45–55% operating cost ratio. D3 RIN revenue stress-tested at \$1.50/gal floor. Tire BOO spread is net after City rent. All figures require independent feasibility analysis before investor or lender presentation.

SECTION 2 — LAND CONTROL: THE GROUND LEASE STACK

City/DCIDA → M4CleanTech → Operators: No Acquisition Capital

M4CleanTech does not need to purchase land to control it. A long-term City ground lease provides the legal position required to sublease to industrial operators, attract grant capital, and build equity — without acquisition cost or personal guarantees. The ground lease is simultaneously M4CleanTech's real estate instrument, its procurement credential, and its grant application anchor.

TIER 1 — LAND OWNER	TIER 2 — GROUND LESSEE	TIER 3 — OPERATORS
City of Chester / DCIDA Retains fee simple ownership. Land stays public — politically clean for Mayor Roots. Ground lease term: 40–99 years. Nominal or below-market rent to M4CT. City benefit: jobs + tax ratables + brownfield activation + EJ narrative	M4CleanTech LLC Holds the ground lease. Develops both sites. Procures grants. Subleases to AD/RNG operator on Parcel A and to Klean/KBR on Parcel B. Pockets the spread on both. No acquisition capital. No personal guarantee. Revenue: both spreads + developer fee + equity stakes	AD/RNG Operator (Parcel A) + Klean/KBR BOO (Parcel B) AD/RNG operator (Bioenergy Devco or equivalent) develops and operates the RNG facility. Klean/KBR builds, owns, and operates tire pyrolysis BOO on separate parcel. Both pay ground rent to M4CT. Both fund their own capital. Combined sublease rent: \$200K–\$450K/yr to M4CT before City cost

THE RIGHT OF FIRST REFUSAL MUST BE IN THE GROUND LEASE

When M4CleanTech signs the Phase 1 ground lease with City/DCIDA, the agreement must include a Right of First Refusal on adjacent waterfront parcels — specifically any acreage released by the Reworld closure. This costs nothing to negotiate at signing but protects the site for Phase 2 expansion (second AD train or Ecolomondo-equivalent owned facility at Year 4+). Trevor must raise this as a ground lease term before signing.

SECTION 3 — AD/RNG DEVELOPMENT PARTNERS

Who Builds and Operates the AD/RNG Facility

M4CleanTech is the site developer and ground lessor — not the AD/RNG operator. The operational model mirrors the BOO structure used for tire pyrolysis: M4CleanTech controls the site via ground

lease, brings in a proven AD/RNG developer/operator, and earns developer fee + ground lease spread + potential equity participation. Three target partners are identified below.

PARTNER	WHY THEY FIT CHESTER	PRIORITY	OUTREACH
Bioenergy Devco	Mid-Atlantic AD/RNG developer with operating facility in Jessup, MD. Their model matches M4CT exactly: develop, operate, and offtake on developer-controlled brownfield sites. Jessup facility is the direct precedent for Chester.	Priority Target	Contact: business development via bioenergydevo.com
RNG Energy Solutions	Developed the \$440M Linden NJ AD/RNG facility — the direct comparable to Chester. COO Andy Shea has publicly stated the need for municipal support and site control. Chester ground lease is exactly what they look for.	Priority Target	Contact via ENR Linden project coverage
Vanguard Renewables	Northeast US AD/RNG developer with multiple operating facilities. Focus on food waste and FOG. Have a farm-centric model that adapts to urban brownfield with the right site control structure.	Secondary Target	Contact: projects@vanguardrenewables.com
Clean Earth Capital	Impact-focused AD/RNG developer active in PA and NJ. Smaller firm — potentially more flexible on EJ community benefit terms than larger operators. Good fit for Chester's community benefit agreement requirements.	Secondary Target	Research current deal pipeline before outreach

COMMUNITY BENEFIT AGREEMENT — NON-NEGOTIABLE

Before any AD/RNG operator agreement is signed, M4CleanTech must execute a Community Benefit Agreement with CRCQL (Chester Residents Concerned for Quality Living) — the organization that fought Covanta/Reworld for 30 years. CBA terms: local hiring preference (minimum 40% Chester residents), air quality monitoring with community access to data, annual community health fund contribution (ROM \$50K–\$100K/yr from AD/RNG revenue), and CRCQL representative on M4CleanTech community advisory board. Without CRCQL support, no Chester project moves forward regardless of technology.

SECTION 4 — TIRE BOO: KLEAN/KBR TYRENOVA SECONDARY REVENUE

Tire Pyrolysis BOO: Capital-Free Upside, Not Load-Bearing

The tire BOO component is retained in Chester's strategy but repositioned as secondary revenue. Klean/KBR's TyreNova BOO structure means they build, own, and operate the facility on a subleased parcel — M4CleanTech provides site access and pockets the ground lease spread with zero capital at risk. The AD/RNG facility is fully viable without the tire BOO. The tire BOO is additional upside.

ELEMENT	DESCRIPTION	M4CT POSITION	KEY REQUIREMENT
Why TDP Not Plastics Pyrolysis	Tire feedstock is homogeneous — the core failure mode of plastics pyrolysis (heterogeneous feedstock creating inconsistent yields and contamination) does not apply. Recovered carbon black from tires has an established commodity market. Brightmark/New Hope failures were plastics-specific.	Tire TDP defensible	Klean/KBR has non-US operating references
Klean/KBR BOO Structure	Klean licenses TyreNova on build, own, operate basis with full performance guarantees and flexible project financing. KBR is a \$7B+ global engineering firm. BOO removes facility CapEx requirement from M4CleanTech entirely.	Zero M4CT capital	KBR institutional credibility
Site Requirement	Separate parcel from AD/RNG facility. 3–5 acres minimum. Adjacent waterfront parcel on Reworld site. Right of First Refusal in ground lease secures this parcel option.	3–5 acres	ROFR in ground lease
Revenue to M4CleanTech	Ground lease spread: City nominal rent (\$1–5K/yr) vs. Klean/KBR market industrial rent (\$150–300K/yr). M4CT pockets difference. Plus 5–15% equity stake in BOO facility negotiated at deal formation.	\$150K–\$300K/yr + equity	Day 1 of BOO operations
Risk Position	If Klean/KBR does not proceed, M4CleanTech loses nothing — the parcel is returned under the ground lease sublease. AD/RNG continues carrying the full financial thesis. BOO is upside, not dependency.	Zero downside to M4CT	AD/RNG is load-bearing
Community Framing	Do NOT call this a pyrolysis or thermal facility in public communications. Reference as "clean circular economy	Framing critical	CRCQL pre-clearance required

ELEMENT	DESCRIPTION	M4CT POSITION	KEY REQUIREMENT
	processing" or "tire recovery facility" and ensure CRCQL review of air quality permitting before any announcement.		

<https://kleanindustries.com/about/sales-partnerships/commercial-arrangement/>

SECTION 5 — M4CLEANTECH CHESTER FINANCE STACK

Grant-Heavy. BOO-Structured. M4D Exposure Under \$300K.

TRANCHE	SOURCE & RATIONALE	ROM AMOUNT	COST	STRUCTURE
EPA Brownfields (Site)	Waterfront site Phase I/II ESA + remediation. Chester EJ status maximizes eligibility.	\$500K–\$1.5M	0% grant	Non-dilutive
EPA EJ Thriving Communities	Food waste diversion in Chester EJ community is strongest possible EJ Thriving Communities application.	\$1M–\$3M	0% grant	Non-dilutive
EDA Economic Adjustment Assistance	Chester EJ + distressed community + job creation = strong EDA EAA application. File alongside Chester brief submission.	\$1M–\$3M	0% grant	Non-dilutive
PA DCED RACP	Redevelopment Assistance Capital Program. AD/RNG infrastructure on remediated industrial site = RACP core use case.	\$1M–\$3M	0% grant	Non-dilutive
CDFI Senior Debt — AD/RNG	Tipping contracts + D3 RIN revenue + digestate offtake = bankable project cash flow. Target: Reinvestment Fund, NCALL Research, or PA CDFI.	\$4M–\$8M	4–6%	15–20 yr
AD/RNG Operator Co-Investment	AD/RNG development partner (Bioenergy Devco / RNG Energy Solutions) brings operator equity and construction financing under development agreement with M4CT.	\$6M–\$12M operator	Operator equity	No M4CT capital
Klean/KBR BOO (Tire Facility)	Klean/KBR funds, builds, owns, operates tire BOO on sublease. Zero	\$8M–\$15M operator	BOO operator	No M4CT capital

TRANCHE	SOURCE & RATIONALE	ROM AMOUNT	COST	STRUCTURE
	M4CT capital for this component regardless of outcome.			
M4CleanTech Equity	Predevelopment costs: M4Di FeasOps (\$22K), legal (ground lease, PSA), site work. Covered by developer fee at financial close. Net M4CT capital below \$300K.	<\$300K	Own capital	Recovered at fin. close
TOTAL — M4CleanTech Chester	AD/RNG + Tire BOO combined. M4CT capital exposure below \$300K. Full facility capital from operator + grant stack.	\$21.5M–\$45.5M	Blended 2–4%	M4CT exp: <\$300K

ROM ±30–40%. AD/RNG operator co-investment terms depend on development agreement negotiation. CDFI loan subject to lender underwriting at time of application. Grant amounts competitive and timing-dependent. PA DCED RACP requires job creation commitments.

SECTION 6 — CRITICAL PATH MILESTONES

Sequencing Matters. Each Gate Unlocks the Next.

MONTH 1–3 Site Control + Feedstock LOIs Gates: EDA filing + CRCQL CBA Execute EDA with City/DCIDA on Reworld waterfront. Initiate CRCQL CBA negotiation. File M4Di FeasOps P3 engagement. Contact Bioenergy Devco and RNG Energy Solutions for AD/RNG site feasibility. Initiate feedstock LOIs from regional haulers and FOG processors. Contact Klean/KBR for TyreNova BOO site feasibility on Parcel B.	MONTH 3–9 PA DEP Pre-Application + Feedstock Contracts Gates: CDFI pre-qual + grant filing PA DEP pre-application meeting for AD/RNG air quality + solid waste permits. Execute feedstock purchase agreements from LOIs. File EPA EJ + EDA EAA grant applications with Mayor endorsement letters. CDFI pre-qualification with Reinvestment Fund. Pipeline interconnection pre-application with Enbridge/SJI.	MONTH 9–18 Permits + Financial Close Prep Gates: operator agreement + CDFI commitment PA DEP solid waste and air quality permits for AD/RNG. AD/RNG operator development agreement executed. CDFI loan commitment letter. Klean/KBR BOO sublease executed (if proceeding). M4CT developer fee agreement signed. Community Benefit Agreement with CRCQL finalized.	MONTH 18–36+ Construction + Operations Gates: Year 2 revenue online AD/RNG facility construction (12–18 month build). Pipeline interconnection commissioned. First D3 RINs generated. Tipping fees flowing. BOO facility construction (Klean/KBR funded). Ground lease spreads flowing from both operators. M4CT annual recurring revenue: \$3.65M–\$7.2M/yr gross.
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All ROM investment, revenue, and savings figures are preliminary estimates with $\pm 30\text{--}40\%$ confidence range. Not guaranteed, not funded commitments, and not GMP. Produced for concept-level conversation and to frame the scope of M4Di predevelopment engagement. Revenue figures are gross before operating expenses. AD/RNG revenue depends on D3 RIN pricing, tipping fee contracts, and pipeline interconnection agreement terms — all require negotiated execution before financial close. Ground lease, BOO sublease, and supplier exclusivity arrangements require legal review. Grant capital is subject to competitive application and lender underwriting. Pennsylvania regulatory compliance requires PA-licensed counsel. DNREC environmental review and DEDO approvals are conditions precedent to any DE construction commitment.